

Late Onset Postpartum Preeclampsia

Section 1: Case Summary

Scenario Title:	Late Onset Postpartum Preeclampsia
Keywords:	Preeclampsia, hypertension, seizure
Brief Description of Case:	A G ₁ T ₁ A ₀ 20-year-old female 5 weeks postpartum after a cesarean section is brought in by ambulance with worsening upper abdominal pain onset 1 week ago, 2 days of worsening dyspnea and a new severe headache. The patient is found to be increasingly hypertensive and eventually has a seizure. As the patient's condition deteriorates the team must determine the cause of this patient's symptoms (pre-eclampsia) and aggressively treatment her symptoms.

Goals and Objectives	
Educational Goal:	Manage the acute care of late onset postpartum pre-eclampsia.
Objectives: (Medical and CRM)	• Verbalize broad differential diagnosis of abdominal pain and dyspnea in a postpartum patient • Diagnose postpartum preeclampsia through clinical diagnosis • Appropriately and aggressively treat preeclampsia, hypertension and seizure • Identify a decompensating patient with postpartum preeclampsia • Manage the airway in hypoxic seizing patient • Select appropriate specialists for consultation • Effectively communicate patient condition and initiate transfer of care • Maintain adequate communication with patient and her partner
EPAs Assessed:	

Learners, Setting and Personnel					
Target Learners:	☐ Junior Learners		☒ Senior Learners		☐ Staff
	☒ Physicians		☐ Nurses	☐ RTs	☐ Inter-professional
	☐ Other Learners:				
Location:	☒ Sim Lab		☐ In Situ		☐ Other:
Recommended Number of Facilitators:	Instructors: 1				
	Sim Actors: 1				
	Sim Techs: 1				

Scenario Development	
Date of Development:	28/10/2020
Scenario Developer(s):	Katerina Meassick MS-IV, Steven Butler PhD, Shani Italiya DO
Affiliations/Institutions(s):	Christus Health – Texas A&M, Arizona College of Osteopathic Medicine
Contact E-mail:	meassickkaterina@gmail.com
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Section 2A: Initial Patient Information

A. Patient Chart					
Patient Name: Alice Jones		Age: 20	Gender: Female	Ht: 5'2"/157cm	Wt: 80kg
Presenting complaint: Shortness of breath and upper abdominal pain					
Temp: 36.5°C	HR: 126	BP: 170/105	RR: 22	O ₂ Sat: 92%	FiO ₂ : RA
Cap glucose: 5.6			GCS: 15 (E4 V5 M6)		
Triage note: A 20-year-old female with no significant past medical history is brought in by ambulance with worsening upper abdominal pain onset 1 week ago when she woke up. She has felt nauseous and has vomited one time this morning. Two days ago, she began to feel short of breath. She states that it has been getting worse and she is now having trouble lying flat. She was a little hypertensive with EMS. They placed one 20 gauge IV in the left AC.					
Allergies: NKDA					
Past Medical History: None			Current Medications: Prenatal vitamins No herbal, OTC medication		
Past Surgical History: Elective Cesarean section - 5 weeks ago, no complications			Social History: No tobacco, alcohol or recreational drug use Drinks one cup coffee a day She is a student at a local university		

Section 2B: Extra Patient Information

A. Further History	
<u>Partner:</u> Has always been healthy. The only thing her doctor told her before she got pregnant is that she should try to lose some weight. She had the baby 5 weeks ago via cesarean section. Scheduled to see OB next week.	
<u>Patient</u> • The patient has a headache that started this morning. Sharp, bilateral, with photophobia. No thunderclap. • The patient feels a short of breath. No cough, no chest pain, no phlegm production. Worse on exertion. • The patient's abdominal pain is in her epigastric and right upper quadrant. Progressive over one week, now is a 9/10, constant and dull. Increasingly nauseous and has vomited one time today. No bowel changes. • Urine has been bubbly and darker for the past week. No dysuria, burning, or itching. • Hands and legs have been swollen for the past week. She has gained 5lbs in the past week. • 5 weeks postpartum after an elective cesarean section. Unremarkable pregnancy. No vaginal bleeding or discharge.	
B. Physical Exam	
Head & Neck: Nil acute	Gyn: Pelvic exam normal (if done)
Cardio: Tachycardic. Pitting edema to bilateral feet and hands.	Neuro: Normal mental status. Normal strength/sensation. Mildly hyper-reflexic
Resp: Bilateral diffuse crackles. Increased work breathing but patient still able to speak in full sentences.	MSK/skin: No rashes, lesions, jaundice.
Abdo: Soft, RUQ tenderness. No suprapubic tenderness.	Other: nil



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Section 3: Technical Requirements/Room Vision

A. Patient
☒ Mannequin (<i>adult</i>) High-fidelity manikin simulator
☐ Standardized Patient
☐ Task Trainer
☐ Hybrid
B. Special Equipment Required
Airway Cart including RSI Supplies Cardiorespiratory Monitor Crash Cart
C. Required Medications
For Preeclampsia: IV Magnesium sulfate 1 st Line Anti-Hypertensive: IV Labetalol, IV Hydralazine, PO Nifedipine 2 nd Line Anti-Hypertensive: IV Nitroglycerine, IV Nitroprusside 1 st Line Anti-Seizure Meds: IV Lorazepam, IV Diazepam 2 nd Line Anti-Seizure Meds: IV Phenytoin RSI Medications
D. Moulage
Female appearing wig Patient gown
E. Monitors at Case Onset
☐ Patient on monitor with vitals displayed ☒ Patient not yet on monitor
F. Patient Reactions and Exam
<i>Include any relevant physical exam findings that require mannequin programming or cues from patient (e.g. – abnormal breath sounds, moaning when RUQ palpated, etc.) May be helpful to frame in ABCDE format.</i> A: Patent, patient is speaking clearly B: Equal bilateral air entry, crackles heard bilaterally, tachypneic C: Tachycardic, normal pulses, RUQ tenderness D: GCS 15 (E4, V5, M6), pupils 2mm reactive, photophobic E: No signs of trauma or bleeding

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Section 4: Sim Actors and Standardized Patients

Sim Actors and Standardized Patient Roles and Scripts	
Role	Description of role, expected behavior, and key moments to intervene/prompt learners. Include any script required (including conveying patient information if patient is unable)
Patient's Partner	Anxious but helpful. Knows patient's full history and is forthcoming with information. Asks periodically for updates.
Bedside Nurse	Skilled and helpful critical care ED nurse. Will prompt to various aspects of clinical care if team leader has missed something.

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Section 5: Scenario Progression

Scenario States, Modifiers and Triggers				
Patient State/Vitals	Patient Status	Learner Actions, Modifiers & Triggers to Move to Next State		Facilitator Notes
1. Baseline State Rhythm: Sinus HR: 126 BP: 170/105 RR: 23 O ₂ SAT: 92% (RA) T: 98.9°C GCS: 15	Patient is alert and oriented, in distress. Currently have abdominal pain, dyspnea, severe headache.	<u>Expected Learner Actions</u> ☐ Cardioresp monitors ☐ Supplemental oxygen therapy ☐ Interview patient ☐ Preform physical exam ☐ Order 2 nd IV and bloodwork ☐ Start loading dose Mg sulfate ☐ Start 1st line antihypertensive ☐ Order chest x-ray ☐ Verbalize broad differential diagnosis	<u>Modifiers</u> <i>Changes to patient condition based on learner action</i> - Place nasal prongs → O ₂ SAT 95% - At 3 min, nurse prompts to recent delivery <u>Triggers</u> <i>For progression to next state</i> - 5 minutes or all actions complete	
2. Decline Rhythm: Sinus HR: 120 BP: 165/95 RR: 22 O ₂ SAT: 95% (NP) GCS: 10 (E3V2M5)	Patient is becoming somnolent. She cannot answer questions anymore. Her eyes are half closed.	<u>Expected Learner Actions</u> ☐ Re-check vitals after treatment ☐ Give second antihypertensive agent ☐ Check cap glucose	<u>Modifiers</u> - Place nasal prongs → O ₂ SAT 95% - Nurse prompts to high BP <u>Triggers</u> - 2 nd antihypertensive agent given	
3. Seizure Rhythm: Sinus HR: 140 BP: 170/110 RR: 24 O ₂ SAT: 88% GCS: Active seizure	Patient begins seizing.	<u>Expected Learner Actions</u> ☐ Check cap glucose ☐ Re-bolus magnesium sulfate ☐ Give benzodiazepine ☐ O ₂ by NRB ☐ Consider need for intubation	<u>Modifiers</u> - Give first line seizure medication → does not stop seizure <u>Triggers</u> - All action completed	



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4. Ongoing Seizure Rhythm: Sinus HR: 140 BP: 175/115 RR: 24 O ₂ SAT: 82% GCS: Active seizure	Patient continues seizure.	<u>Expected Learner Actions</u> ☐ Consider predictors of difficult airway ☐ Neuroprotective RSI ☐ Intubate patient ☐ Give 2 nd line anti-seizure medication	<u>Modifiers</u> - Nurse to cue learner to hypoxia <u>Triggers</u> - Pt intubated	
5. Stabilized State Rhythm: Sinus HR: 100 BP: 120/65 RR: 12 O ₂ SAT: 100 % GCS: 2T (paralyzed)	Patient is stable and intubated.	<u>Expected Learner Actions</u> ☐ Routine post-intubation care ☐ Admit patient to ICU ☐ Consult OB/GYN ☐ Consult neurology ☐ Order CT Head ☐ Update Family	<u>Modifiers</u> - Nurse prompts to consultant services - If learner fails to update family → husband asks for an update <u>Triggers</u> - Consulting neurology AND OB/GYN, admitting to ICU and updating family ends the case	

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Appendix A: Laboratory Results

<u>Urine Dip</u> Appearance – Brown, bubbly Specific gravity – 1.023 pH – 6.0 Glucose – negative Bilirubin – negative Ketone – Trace Hemoglobin – Negative Protein- 3+ Nitrite - Negative Leukocytes - negative	<u>Other</u> B-HCG - negative
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Appendix B: ECGs, X-rays, Ultrasounds and Pictures

Paste in any auxiliary files required for running the session. Don't forget to include their source so you can find them later!



<https://www.semanticscholar.org/paper/Late-postpartum-preeclampsia-induced-acute-edema%2C-a-Rua-Costa/06f9edc1fc8a1a569b5e77f41f4cd6b0ba3bca30>

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Appendix C: Facilitator Cheat Sheet & Debriefing Tips

Include key errors to watch for and common challenges with the case. List issues expected to be part of the debriefing discussion. Supplemental information regarding any relevant pathophysiology, guidelines, or management information that may be reviewed during debriefing should be provided for facilitators to have as a reference.

Key moments to reflect on (extensive list, choose appropriate focus for individual learners):

- Clinical assessment and physical exam in narrowing the broad differential diagnosis of abdominal pain
- Importance of understanding risk factors for late onset postpartum preeclampsia
- Appropriate initial management of late onset postpartum preeclampsia
- Appropriate management of a seizure in preeclampsia
- Reassessment of patient including repeat vital signs and reaction to treatment
- Initiation of second and third line antiseizure and antihypertensive treatments while monitoring closely for airway and seizure intervention
- Response to deterioration of patient
- Decisions to begin airway management in a seizing patient
- Importance of treating seizures after paralytics given during intubation
- Importance of maintaining adequate communication with the patient and her partner
- Importance of involving OB/GYN
- Understanding when to involve neurology
- Understanding when a preeclamptic patient needs to be admitted versus when they can be discharged with outpatient follow up

Possible Debriefing Discussion Questions (Focus on process, not knowledge):

- What do you feel went well during this case?
- What aspects of this case were challenging and how did you overcome them?
- What is the differential diagnosis for this patient's specific presentation?
- Do you think your differential was altered because the patient was not currently pregnant?
- How do you think your initial assessment and differential would have been different if this patient had presented while she was still pregnant?
- Do you feel that there is a difference in pressure when caring for a pregnant patient as opposed to a postpartum patient?
- What aspects of the patient's history did you consider when evaluating for preeclampsia?
- How were you able to rule out primary GI and pulmonary pathology?
- When the patient did not initially respond to treatment, did that change your confidence in your diagnosis?
- Why do you give both antihypertensive agents and magnesium sulfate to preeclamptic patients?
- What considerations did you have when you decided to intubate the patient?
- When do you intubate a seizing patient?
- Why is magnesium important in preeclamptic versus eclamptic patients?
- How did your management of this simulated patient compare to actual patients you have had?
- If the patient had responded to initial antihypertensive and antiseizure treatments, how would this have changed your management?
- Would you have chosen different medications if this patient had still been pregnant?



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Medication Summary:

Anti-Hypertensives

- Magnesium sulfate - 4-6 g loading dose over 15-20 min followed by 1-2 g/hour continuous infusion OR 10 g loading dose IM as 5 g IM in each buttock then 5 g every 4 hrs.
- Labetalol - 20 mg IV over 2 min followed by 1-2mg/min titrated to response. Single max dose 80mg, max cumulative dose 300 mg. OR 200 mg PO every 12 hrs, dose may be increased up to 800 mg every 8-12 as needed. Max 2.4 g/day
- Hydralazine - 5 or 10 mg IV infused over 1-2 min with 5-10 mg bolus given after 20 min if no response.
- Nifedipine – Immediate release 10 or 20 mg tablet or 10 mg capsule PO recheck after 20 min and give 10-20 mg orally, If remains above target after 40 min give another 10-20 mg. After third dose consider another class. Rapid-acting or extended-release 30 mg PO.
- Nitroglycerine 5 mcg/minute increased every 3-5 min with a max of 100 mcg/min
- Nitroprusside 0.25-10 mcg/kg/min IV infusion not exceeding 2 mcg/kg/min

Anti-Seizure Meds

- Magnesium sulfate - 4-6 g loading dose over 15-20 min followed by 1-2 g/hour continuous infusion OR 10 g loading dose IM as 5 g IM in each buttock then 5 g every 4 hrs
- Lorazepam - 2-4 mg IV over 2-5 minutes or 0.02-0.03 mg/kg IV max rate of 2mg/min.
- Diazepam - 5-10 mg IV as single dose at 5mg/min may repeat at 3-5 min intervals up to 30mg. Status epilepticus 5-10 mg single dose at 5 mg/min may repeat every 3-5 min
- Phenytoin –IV 15 mg/kg loading dose, max dose 1-1.5 g. IV maintenance 4-7 mg/kg/day

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